

SambaRack SN40-16

AI infrastructure for agentic and generative inference

SambaRack

Rapid advancements in AI are causing an explosion in usage. SambaRack, built with unique SN40L Reconfigurable Dataflow Units (RDUs), delivers the high performance, throughput and ultra-low latency essential for token-hungry agentic and generative applications all within a compact, energy-efficient system. Powering SambaCloud, SambaStack, and SambaManaged, SambaRack is a proven platform for delivering the premium tokens production AI applications and users demand.

Performance at Scale

Run the largest open source models with low latency and high throughput

Use in Existing Data Centers

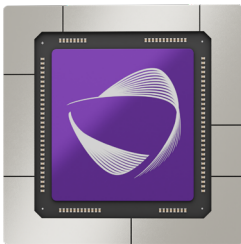
Deploy in today's data centers with an average power consumption of 10kW and air cooling

Deployment Flexibility

With a compact footprint, SambaRack can be used on-premises and even in air-gapped environments

Dataflow delivers efficiency and performance

AI inference is bottlenecked by data movement, not compute speed. SambaNova RDUs leverage dataflow computing to deliver the performance inference at scale demands. With RDUs, you can run the largest models with fewer chips and less infrastructure than alternative inference solutions.



In SambaNova's architecture, compute and memory operate in parallel on-chip. The result is not just more compute, but greater continuity between operations. This is the key distinction: Dataflow isn't simply faster hardware. It's about reducing handoffs, fusing work where possible and keeping the processor fed with the right data at the right time. The result is better performance, lower latency, and reduced energy consumption.

Three tier memory for large models and variety

Real-world inference applications require a variety of models for different tasks. To meet this need, SambaNova RDUs have a three-tier memory hierarchy, each tier serving a specific purpose. On-RDU SRAM handles the hottest local data, HBM streams model weights at scale, and directly attached DDR supports prompt caching and a model catalog for varying needs. High speed model switching enables agentic and reasoning applications to use the best model for each step of a task - avoiding the excessive delays common in other architectures and eliminating the need to deploy separate systems for each model. This enables delivering more powerful inference services with less hardware.

Simple to deploy at scale

Deploying inference infrastructure at scale requires more than just the latest AI processors. SambaRack's low power consumption and air-cooling ensure compatibility with a broad spectrum of data centers. It ships fully prebuilt, cabled, and ready to deploy, helping you get to production faster.

Key Features

- High Performance**
 Industry-leading open-source model token rates and latency
- High Throughput**
 Supports high user counts for better tokenomics
- Low Power Consumption**
 Industry-leading efficiency at 10kW typical per air-cooled rack



SAMBARACK SN40 SPECIFICATIONS

SN40L RDUs	16x SN40™ Reconfigurable Dataflow Units (RDUs), with 520MB SRAM, 64GB HBM, 256GB DDR each
Processing Power	10.2 PFLOPs @BF16
Total RDU SRAM	8GB
Total RDU HBM	1TB
Total RDU attached DDR	4TB
Host processor	2x 64 core CPUs, 2TB DDR4 memory
Host boot and storage	4x 960GB (2x RAID 1 + 2 hot-spares)
Host storage	6x 7.6TB NVMe disks RAID 10 (21TB usable)
Networking	High performance 400/200 GbE data switch
Management	1 GbE switch, Serial console server
Software	Red Hat Enterprise Linux OS
System Dimensions	Height: 78.5" (1994mm) Width: 23.6" (600mm) Depth: 50" (1270mm)
System Weight	1146lbs (520kg)
Power consumption	Inference: 10kW Typical, 16.4kW Max
Airflow requirements	2600 CFM Front to back
Cooling at Max Utilization	56000 BTU/Hr
Operating Temperature	59° F to 86° F (15 C to 30° C)
Operating Humidity	20% to 80% (non-condensing)
Operating Altitude	Up to 9842ft (3000m); derated by @ 1.8F (1C) per 984ft (300m) above 2952ft

CORE SERVICES RACK SPECIFICATIONS

System Dimensions	Height: 78.5" (1994mm) Width: 23.6" (600mm) Depth: 50" (1270mm)
System Weight	922lbs (419kg)
Power consumption	4.8kW Typical, 7kW Max
Airflow requirements	1003 CFM Front to back
Cooling at Max Utilization	24000 BTU/Hr
Operating Temperature	59° F to 86° F (15 C to 30° C)
Operating Humidity	20% to 80% (non-condensing)
Operating Altitude	Up to 9842ft (3000m); derated by @ 1.8F (1C) per 984ft (300m) above 2952ft

Schedule a Meeting → www.sambanova.ai/contact • [Linkedin.com/company/sambanova](https://www.linkedin.com/company/sambanova) • [@SambaNovaAI](https://twitter.com/SambaNovaAI) • info@sambanova.ai



SambaNova is a leader in next-generation AI infrastructure, providing a full-stack platform that delivers the fastest and most efficient AI inference for enterprises, NeoClouds, AI research labs, service providers, and sovereign AI initiatives worldwide. Founded in 2017 and headquartered in San Jose, California, SambaNova offers chips, systems, and cloud services that enable customers to deploy state-of-the-art models with superior performance, lower total cost of ownership, and faster time to value.